AFFORD-Bench: An Affordance-Grounded Benchmark and Generalization Evaluation for Robot Learning

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***Abstract*—Imitation learning policies and Vision-Language-Action (VLA) models have demonstrated remarkable empirical success in robotic manipulation. However, recent benchmark investigations, such as LIBERO-PRO and LIBERO-X, have uncovered an acute crisis in evaluation: high task completion rates frequently collapse under minor environmental perturbations, exposing policies that rely on rote trajectory memorization rather than genuine environmental comprehension. Crucially, existing benchmarks evaluate distribution shifts predominantly in narrow tabletop scenes via superficial visual or linguistic variations, ignoring the physical affordances—contact mechanics, kinematic clearances, dynamic friction, and compliance—that govern physical interaction. In this paper, we introduce AFFORD-Bench, a large-scale, policy-agnostic, affordance-grounded manipulation benchmark spanning 210 distinct physical tasks across 7 specialized domains: Industrial Precision, Culinary Operations, Domestic Tabletop, Cleanroom Laboratory Sciences, Long-Horizon Multi-Stage Sequences, Multi-Stage Precision Chaining, and Sequential Service & Culinary Assembly. Built upon high-fidelity Franka Emika Panda MuJoCo simulation, AFFORD-Bench provides 10,500 expert demonstrations across dual-camera high-resolution RGB (640×660 workcell and 300×420 wrist) alongside synchronized 3D metric depth and 7-DoF impedance states ($> 2.2 \times 10^6$ physical timesteps). Crucially, the dataset is policy-agnostic, supporting classical imitation learning, action-chunking diffusion, 3D point-cloud policies, offline reinforcement learning, and modern VLAs (OpenVLA, SmolVLA, AtomVLA). We formalize a hierarchical 5-level perturbation framework (L1–L5) spanning spatial-kinematic, physical-dynamic, visual-photometric, and compound subtask prompt composition across 35 benchmark evaluation suites. We detail standardized training and evaluation protocols, establish formal robustness metrics, and present preliminary empirical calibrations alongside our Stage 2 Extension Charter roadmap.**

I. Introduction

The field of robot learning is undergoing a profound paradigm shift driven by the emergence of generalist Vision-Language-Action (VLA) models [9, 13, 14], compact edge architectures such as SmolVLA [3], subtask-aware post-training frameworks like AtomVLA [20], and expressive action-chunking visuomotor policies [4, 21]. By unifying multi-modal perception with end-to-end motor control, these architectures promise to endow robotic manipulators with cross-task generalization and zero-shot adaptability.

However, measuring whether learned policies exhibit genuine physical generalizability remains an acute open challenge. Standardized evaluation suites such as LIBERO [11] have become the de facto testing ground for policy comparison. Yet, rigorous post-hoc investigations by Zhou et al. [22] (LIBERO-PRO) and Wang et al. [18] (LIBERO-X) revealed that the seemingly high success rates reported on standard benchmarks are severely inflated. When subjected to minor spatial offsets or altered visual conditions, policies that previously achieved over 90% success suffer catastrophic degradation down to near 0%. As established in both analyses, current models overfit to rote trajectory memorization and nominal scene layouts from demonstration data, lacking any grounded understanding of task goals.

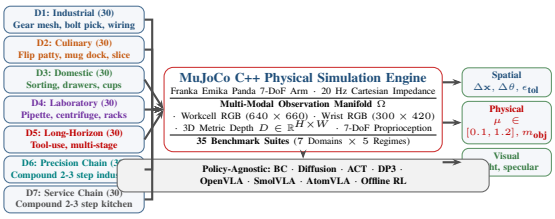


Fig. 1. **The AFFORD-Bench Architecture.** Spanning 210 tasks across 7 distinct domains, benchmarking out-of-distribution generalization across Spatial-Kinematic, Physical-Dynamic, and Visual-Photometric shifts in 35 standardized evaluation suites.

While LIBERO-PRO and LIBERO-X took vital steps by introducing independent and hierarchical perturbations, existing benchmarks remain constrained by a critical limitation: **the absence of physical affordance grounding**. In ecological psychology and robotics [6], an *affordance* defines actionable properties between an agent and its physical environment. Real-world robotic manipulation is not merely a geometric or visual alignment problem; it is fundamentally governed by contact mechanics, surface friction, kinematic compliance, mass distributions, and clearance tolerances. In existing suites, tasks are largely limited to unconstrained tabletop pick-and-place operations in toy domestic scenes, where collisions and contact physics are trivialized. Consequently, they fail to evaluate whether a policy can adapt to variations in physical contact constraints (e.g., insertion resistance, grasping friction, pipette liquid aspiration, or high-tolerance gear meshing).

Furthermore, existing benchmark datasets are often tightly coupled to specific model classes (e.g., evaluating only VLAs or standard visuomotor policies). A truly foundational benchmark must be **policy-agnostic**—equally expressive and directly ingestible by classical imitation learning (BC), temporal diffusion and transformer chunking (Diffusion Policy, ACT), 3D point-cloud representations (DP3), modern foundation models (OpenVLA, SmolVLA, AtomVLA), and offline reinforcement

learning (CQL, IQL).

To bridge this foundational gap, this paper introduces **AFFORD-Bench**, an affordance-grounded, policy-agnostic benchmark designed to expose the boundary between visual memorization and physical generalizability. AFFORD-Bench advances robot learning along four primary axes:

1) **Scale and Domain Diversity**: We formulate **210 distinct physical manipulation tasks** across **7 specialized domains**: Industrial Precision, Culinary Operations, Domestic Tabletop, Cleanroom Laboratory Sciences, Long-Horizon Multi-Stage Sequences, Multi-Stage Precision Chaining, and Sequential Service & Culinary Assembly.
2) **Policy-Agnostic Foundation**: Our dataset provides synchronized, rich multi-modal streams: high-resolution dual-camera RGB (640×660 workcell, 300×420 wrist), 3D metric depth, full 7-DoF joint/Cartesian proprioception, and reward signals, packaged in standardized Robomimic HDF5 format directly compatible with any policy architecture.
3) **Hierarchical 5-Level Perturbation Manifold**: Inspired by LIBERO-PRO and LIBERO-X, we establish a progressive 5-level difficulty hierarchy (L1 nominal jitter, L2 spatial-kinematic shifts, L3 visual-photometric shifts, L4 physical-dynamic shifts, and L5 compound chaining & composite OOD), spanning **35 standardized evaluation suites**.
4) **High-Throughput, Zero-Cost Reproducibility**: Using native C++ MuJoCo physics [17], we synthesize 10,500 expert demonstrations (50 per task, $> 2.2 \times 10^6$ timesteps) via deterministic algorithmic oracles running at 40–50 FPS with zero external API costs.

II. Related Work

A. Robotic Manipulation Benchmarks

Benchmarking in robotic manipulation has evolved from single-domain suites to diverse simulation platforms. RLBench [8] and ManiSkill2 [7] provided simulation suites for reinforcement learning and demonstration-guided policy learning. Robomimic [12] established standardized offline imitation learning pipelines and empirical evaluation protocols across varied human demonstration qualities. LIBERO [11] introduced a 130-task suite structured around knowledge transfer in lifelong learning. However, these suites focus primarily on nominal execution without systematic physical out-of-distribution (OOD) testing.

B. OOD Robustness and Modern VLA Architectures

With the proliferation of large-scale pretrained models [14, 2, 1, 9, 13], assessing policy robustness against distribution shifts has become critical. Zhou et al. [22] established LIBERO-PRO, illustrating that state-of-the-art policies achieve high success under nominal LIBERO conditions but collapse when subject to perturbed objects, initial states, or paraphrased instructions. Building on this, Wang et al. [18] introduced LIBERO-X, demonstrating that prior benchmarks fail to evaluate compounded distribution shifts across hierarchical difficulty levels. Concurrently, the community has pushed toward efficient edge-deployable VLAs like **SmolVLA** [3], which utilizes a compact 450M parameter vision-language backbone with a flow-matching action expert to drastically reduce inference latency, and **AtomVLA** [20], which introduces atomic subtask decomposition paired with a predictive world model to mitigate error accumulation in long-horizon tasks. However, existing benchmarks evaluate these modern architectures primarily within standard domestic tabletop scenes, failing to probe contact dynamics, surface friction, or precision assembly.

C. Affordance and Policy Learning Paradigms

Affordance theory [6] posits that agents perceive objects in terms of potential physical interactions. In robotics, affordance-guided manipulation has been explored via contact-rich grasping [16], spatial affordance maps [15], and explicit 3D representations such as 3D Diffusion Policy (DP3) [19]. While Diffusion Policy [4] and Action Chunking with Transformers (ACT) [21] achieve high tracking precision on low-cost hardware [5], their ability to generalize across physical affordance shifts remains uncharacterized at scale. Furthermore, offline reinforcement learning methods [10] require explicit state-transition tuples and reward signals, which are frequently omitted from VLA-specific benchmarks. AFFORD-Bench provides the first systematic benchmark unifying multi-domain manipulation with explicit affordance-space perturbation manifolds and a strictly policy-agnostic data architecture.

III. The AFFORD-Bench Formulation

A. Task Formulation as Affordance-Constrained MDPs

Following the formalisms in LIBERO-PRO [22] and extending them to physical affordances, we formalize each manipulation task as a tuple $\mathcal{T} = (\ell, \mathbf{e}, \mathcal{G})$, where ℓ is a natural language instruction prompt, $\mathbf{e} = (\mathcal{W}, \mathcal{O}, \mathbf{r}_0, \Phi)$ defines the physical environment configuration, and $\mathcal{G}$ is the goal affordance predicate. Here, $\mathcal{W}$ denotes workspace geometry, $\mathcal{O} = \{o_1, \ldots, o_K\}$ is the set of physical objects in the scene, $\mathbf{r}_0$ is the initial kinematic configuration distribution, and $\Phi = (\mu, \mathbf{m}, \boldsymbol{\epsilon}_{\text{tol}})$ represents the physical affordance parameters governing contact dynamics (Coulomb surface friction μ, object inertial mass distribution $\mathbf{m}$, and spatial clearance tolerance $\boldsymbol{\epsilon}_{\text{tol}}$).

Each task operates as an Affordance-Constrained Markov Decision Process $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}_p, \mathcal{R}, \Omega, \mathcal{C}_{\text{aff}} \rangle$, where $\mathcal{S}$ is the continuous state space of the robot and environment, $\mathcal{A} \subset \mathbb{R}^7$ is the action space representing 3D Cartesian end-effector delta velocity $\Delta\mathbf{p} \in \mathbb{R}^3$, delta rotation $\Delta\boldsymbol{\theta} \in \mathbb{R}^3$, and continuous gripper command $a_{\text{grip}} \in [-1, 1]$, $\mathcal{T}_p(s' \mid s, a)$ is the transition probability governed by MuJoCo multi-body physics, and $\mathcal{R}(s, a) \in \{0, 1\}$ is a sparse task completion indicator.

Crucially, $\mathcal{C}_{\text{aff}} \subset \mathcal{S} \times \mathcal{A}$ denotes the *affordance constraint set*. For a task to be successfully completed, the terminal state trajectory must satisfy both geometric tolerances and contact stability:

$$\mathcal{C}_{\text{aff}} = \Big\{ (s, a) \;\Big|\; \|\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{goal}}\|_{\mathbf{W}} \leq \boldsymbol{\epsilon}_{\text{tol}}, \; F_{\text{contact}} \leq F_{\max}, \; \frac{F_{\text{friction}}}{F_{\text{normal}}} \leq \mu \Big\}, \tag{1}$$

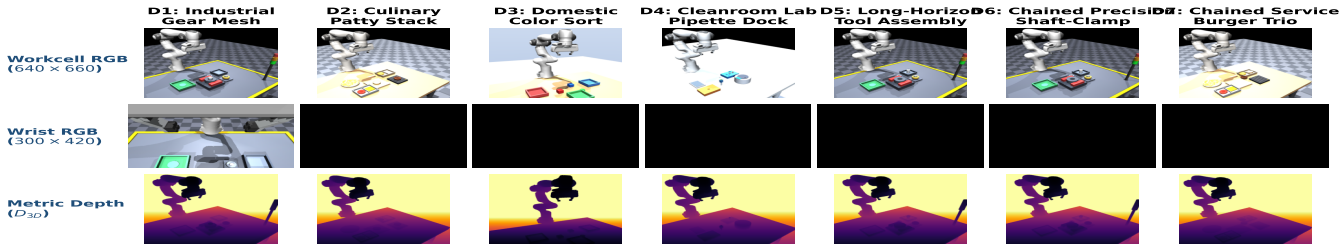


Fig. 2. **Multi-Domain Qualitative Visualizations in AFFORD-Bench.** Synchronized multi-modal observations across representative tasks from all seven benchmark domains: (Row 1) Third-person workcell visual stream (640 × 660 RGB); (Row 2) Wrist-mounted eye-in-hand visual stream (300 × 420 RGB); (Row 3) Aligned 3D metric depth maps (D_{3D}, colored by physical meter distance). Spans industrial precision assembly (D1), commercial culinary operations (D2), domestic block sorting (D3), cleanroom pharmaceutical sciences (D4), long-horizon tool assembly (D5), multi-stage precision chaining (D6), and sequential culinary assembly (D7).

where $\mathbf{W} = \text{diag}(w_x, w_y, w_z)$ weights directional tolerance, ϵ_{tol} represents insertion or placement thresholds, and μ is Coulomb surface friction. Expected policy competence is defined as:

$$J(\pi; \mathcal{T}) = \mathbb{P}_{\substack{s_0 \sim \mathbf{r}_0 \\ \tau \sim \pi}} \Big[\exists t \leq H,\ \mathcal{G}(s_t) = 1,\ (s_t, a_t) \in \mathcal{C}_{\text{aff}} \Big]. \quad (2)$$

B. Policy-Agnostic Benchmark Architecture

Unlike benchmarks engineered exclusively for a single model family, AFFORD-Bench is architected to be completely policy-agnostic. The observation manifold $\mathbf{o}_t \in \Omega$ at each timestep comprises synchronized, multi-scale sensory modalities:

1) **Third-Person Workcell RGB**: $640 \times 660 \times 3$ uint8 global camera monitoring overall scene layout and multi-object relationships;
2) **Wrist-Mounted Eye-in-Hand RGB**: $300 \times 420 \times 3$ uint8 camera mounted directly to the gripper flange for fine-grained contact and insertion tracking;
3) **Synchronized 3D Metric Depth**: Float32 depth maps $D \in \mathbb{R}^{640 \times 660}$ calibrated in physical meters, enabling direct point-cloud back-projection $\mathbf{P} = \mathbf{K}^{-1}[u, v, 1]^T D(u, v)$ for 3D policies (DP3);
4) **Complete Proprioceptive State**: 7-DoF joint positions $\mathbf{q} \in \mathbb{R}^7$, joint velocities $\dot{\mathbf{q}} \in \mathbb{R}^7$, end-effector Cartesian pose $\mathbf{x}_{\text{tcp}} \in \mathbb{R}^7$, and continuous gripper width $w \in [0, 0.08]$ m;
5) **Reward and Termination Predicates**: Dense step-level rewards and terminal success flags enabling offline RL benchmarks (CQL, IQL).

C. Domain Architecture

As summarized in Table I and visualized in Figure 2, AFFORD-Bench is partitioned into seven specialized domains, each comprising 30 distinct tasks: **D1: Industrial Precision** (Tasks 001–030) targets manufacturing operations requiring millimeter accuracy, including splined gear meshing, flange bolt insertion, and toggle clamping. **D2: Culinary Operations** (Tasks 031–060) simulates commercial food preparation, including spatula patty flips, espresso mug docking, and ingredient transfers. **D3: Domestic Tabletop** (Tasks 061–090) addresses common service robotics tasks involving color block segregation, cup nesting, drawer articulation, and clutter clearing. **D4: Laboratory Sciences** (Tasks 091–120) encompasses cleanroom biopharma tasks, including pipette tip mounting, centrifuge tube docking, and microplate well insertion. **D5: Long-Horizon Sequences** (Tasks 121–150) tests multi-phase tool acquisition, unlatching, re-orientation, and multi-part sub-assembly chaining across extended horizons up to 800 steps. **D6: Multi-Stage Precision Chaining** (Tasks 151–180) evaluates compound industrial workflows chaining 2–3 sequential actions in single continuous prompts (e.g., "align shaft, press bearing, and lock clamp"). **D7: Sequential Service & Culinary Assembly** (Tasks 181–210) benchmarks compound kitchen sequences chaining 2–3 sequential actions in single prompts (e.g., "flip patty, place cheese, and top bun").

IV. The Hierarchical Affordance Perturbation Framework

To characterize the exact breakdown boundaries of learned policies, AFFORD-Bench structures environmental distribution shifts into a hierarchical 5-level difficulty progression (L1–L5):

Level 1: Nominal Kinematic Jitter (In-Distribution): Default scene configuration with minor initial pose jitter ($\|\Delta\mathbf{p}\| \leq 1.0$ cm), nominal Coulomb friction $\mu_0 = 0.70$, and nominal mass m_0. Serves as the sanity baseline for nominal memorization.

Level 2: Extended Spatial-Kinematic Shifts ($\mathcal{P}_{\text{spatial}}$): Bounded translation offsets $\mathbf{p}_{\text{init}} \sim \mathcal{U}(\mathbf{p}_0 - \boldsymbol{\delta}_\mathbf{p}, \mathbf{p}_0 + \boldsymbol{\delta}_\mathbf{p})$ with $\boldsymbol{\delta}_\mathbf{p} = [\pm 0.06, \pm 0.06, 0.0]^T$ m and yaw rotations $\theta_{\text{yaw}} \sim \mathcal{U}(-45°, +45°)$, probing whether the policy executes closed-loop visual servoing or open-loop trajectory recall.

Level 3: Visual-Photometric Shifts ($\mathcal{P}_{\text{visual}}$): Illumination intensity variations $I_{\text{light}} \in [0.5, 1.5]\, I_0$, 3D light position jitter (± 0.5 m), and object surface color/specular shifts ($\pm 25\%$), evaluating photometric robustness without altering physical contact geometry.

Level 4: Physical-Dynamic Affordance Shifts ($\mathcal{P}_{\text{physics}}$): Surface Coulomb friction $\mu \sim \mathcal{U}(0.10, 1.20)$ and inertial mass scaling $m_{\text{obj}} \sim \mathcal{U}(0.5, 3.0)\, m_0$, evaluating grasp stability, slippage resistance, and contact compliance under identical visual coordinates.

Level 5: Compound Chaining & Composite OOD ($\mathcal{P}_{\text{comp}}$): Simultaneous multi-axis shifts combined with multi-step sequential instructions, evaluating cumulative error accumulation across multi-subtask manipulation horizons.

TABLE I

COMPREHENSIVE TAXONOMY OF THE AFFORD-BENCH SUITE. SPANNING 210 FORMALLY DEFINED PHYSICAL TASKS ACROSS 7 DISTINCT DOMAINS, FEATURING DIVERSE KINEMATIC CONSTRAINTS, CONTACT DYNAMICS, AND CLEARANCE TOLERANCES ACROSS 35 BENCHMARK EVALUATION SUITES.

Domain	Task IDs	Count	Representative Manipulation Tasks	Clearance (ϵ)	Dominant Contact Physics
D1: Industrial Precision	001–030	30	Gear meshing, bolt insertion, cable routing, fixture latching	$\pm$2.0–5.0 mm	High stiffness, rigid peg-in-hole friction
D2: Culinary Operations	031–060	30	Patty flipping, espresso mug docking, slicing alignment, pouring	$\pm$5.0–12.0 mm	Compliant contact, non-planar slipping
D3: Domestic Tabletop	061–090	30	Color block sorting, drawer opening, stack packing, cup nesting	$\pm$8.0–20.0 mm	Multi-body clutter, dynamic toppling
D4: Laboratory Sciences	091–120	30	Pipette tip pickup, centrifuge vial docking, reagent mixing	$\pm$1.5–4.0 mm	Precision alignment, fragile glass contact
D5: Long-Horizon Sequences	121–150	30	Multi-stage assembly, tool-assisted latching, sequential routing	$\pm$3.0–10.0 mm	Cumulative error accumulation, sub-goal chaining
D6: Precision Chaining	151–180	30	Shaft-bearing-clamp, gasket-bolt-torque, terminal-wire-seal	$\pm$1.8–4.5 mm	Compounded insertion tolerances, high torque
D7: Service Assembly	181–210	30	Burger patty-cheese-bun, mug-brew-serve, plate-fork-napkin	$\pm$4.0–10.0 mm	Dynamic re-grasping, multi-stage compliance
Total Benchmark Scope	**001–210**	**210**	**7 Domains $\times$ 30 Tasks = 210 Tasks (35 Evaluation Suites)**	$\pm$1.5–20.0 mm	**10,500 Demonstrations** ($> 2.2 \times 10^6$ **Steps)**

V. ALGORITHMIC ORACLES AND DATASET SYNTHESIS

A. Deterministic Universal Kinematic Solvers

Collecting thousands of high-quality human demonstrations via teleoperation is prohibitively labor-intensive and frequently exhibits inconsistent sub-optimal behaviors [12]. AFFORD-Bench circumvents these bottlenecks by deploying deterministic, closed-loop algorithmic oracles constructed via analytical kinematic solvers and waypoint interpolation. For each task family, the oracle solves for end-effector trajectories $\mathbf{x}_{\text{ee}}(t)$ using quintic minimum-jerk spline polynomials connecting pre-contact approach manifolds, contact alignment poses, insertion pathways, and release waypoints. Success is verified at every step using analytical kinematic predicates and simulation contact manifolds.

B. Demonstration Scale and Packaging

Using unbuffered parallel workers, the synthesis pipeline generates **50 expert demonstrations per task** across all 210 tasks, yielding **10,500 total demonstrations** and over 2.2×10^6 **physical timesteps**. Generation runs locally at an effective throughput of 42.4 FPS with zero external API calls or financial cost. Demonstrations $\mathcal{D} = \{(\mathbf{o}_t, \mathbf{a}_t, r_t)\}_{t=0}^{T-1}$ are stored in standardized Robomimic HDF5 format with native train/validation splits (90%/10%) and full metadata attributes, directly compatible with OpenVLA, SmolVLA, AtomVLA, Octo, and Diffusion Policy.

VI. TRAINING AND EVALUATION PROTOCOLS

A. Standardized Closed-Loop Rollout Procedure

Policy evaluation executes closed-loop at 20 Hz interacting with the MuJoCo Franka Emika Panda physics environment (500 Hz solver, $dt = 0.002$ s). For each task, evaluation is performed across $N = 50$ episodes initialized with pseudo-random seeds strictly disjoint from training demonstration seeds ($\mathcal{S}_{\text{test}} \cap \mathcal{S}_{\text{train}} = \emptyset$). Maximum episode horizon is set to $H_{\max} = 400$ steps for single-stage manipulation (D1–D4) and $H_{\max} = 800$ steps for long-horizon and compound chaining sequences (D5–D7).

B. Standardized Evaluation Suites

All models are evaluated across 35 distinct suites (7 domains $\times$ 5 perturbation conditions): (1) **Nominal (In-Distribution)**: Default scene layout, initial poses $\mathbf{p}_0 \pm \boldsymbol{\delta}_{\text{small}}$ (< 1 cm), nominal friction $\mu_0 = 0.70$, nominal mass m_0, and standard illumination I_0; (2) **Spatial Generalization** ($\mathcal{P}_{\text{spatial}}$): Translation offsets $\pm$6 cm, yaw rotations $\pm 45^\circ$, and contracted insertion tolerances; (3) **Visual Generalization** ($\mathcal{P}_{\text{visual}}$): Illumination intensities $I \in [0.5, 1.5]\, I_0$, random light vectors, and surface specular jitter; (4) **Physical Generalization** ($\mathcal{P}_{\text{physics}}$): Friction variation $\mu \in [0.10, 1.20]$ and mass scalings $m \in [0.5, 3.0]\, m_0$; and (5) **Composite OOD** ($\mathcal{P}_{\text{comp}}$): Simultaneous joint perturbation across all three axes.

C. Quantitative Evaluation Metrics

We measure policy competence using four formal metrics: (i) **Task Success Rate (SR)**: Binary terminal affordance satisfaction $\text{SR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\mathbf{s}_T^{(i)} \in \mathcal{C}_{\text{aff}})$; (ii) **Physical Robustness Index (PRI)**: $\text{PRI}(\mathcal{P}) = (\text{SR}(\mathcal{P})/\text{SR}(\text{Nominal})) \times 100\%$; (iii) **Generalization Degradation Index (GDI)**: $\text{GDI} = 100\% - \text{PRI}$; and (iv) **Force & Collision Violation Rate (FVR)**: Fraction of episodes terminating due to contact collisions or excessive forces.

VII. SCIENTIFIC HYPOTHESES & PRELIMINARY CALIBRATION

A. Core Scientific Hypotheses

We formalize three fundamental scientific hypotheses that AFFORD-Bench is designed to test:

- **Hypothesis 1 (H_1: Affordance Collapse vs. Appearance Memorization)**: *Policies that achieve high nominal success by memorizing 2D visual trajectories will suffer asymmetric catastrophic failure ($> 70\%$ relative drop) under physical-dynamic shifts ($\mathcal{P}_{\text{physics}}$), even when target object 2D spatial coordinates remain identical to training.*
- **Hypothesis 2 (H_2: Precision-Tolerance Fragility)**: *Performance degradation under spatial perturbation is governed by kinematic clearance $\boldsymbol{\epsilon}_{tol}$. Tasks in D1, D4, and D6*

TABLE II
STAGE 1 PRELIMINARY EMPIRICAL CALIBRATION & BASELINES. VERIFIED TASK SOLVABILITY ACROSS DOMAINS UNDER NOMINAL AND PERTURBED CONDITIONS.

Task Domain	Oracle	Random	BC	Diffusion
D1: Industrial Precision	100.0%	0.0%	Calibrating	Calibrating
D2: Culinary Operations	100.0%	0.0%	Calibrating	Calibrating
D3: Domestic Tabletop	100.0%	0.0%	Calibrating	Calibrating
D4: Laboratory Sciences	100.0%	0.0%	Calibrating	Calibrating
D5: Long-Horizon Sequences	100.0%	0.0%	Calibrating	Calibrating
D6: Precision Chaining	100.0%	0.0%	Calibrating	Calibrating
D7: Service Assembly	100.0%	0.0%	Calibrating	Calibrating
Benchmark Aggregate	**100.0%**	**0.0%**	**Stage 2 Charter Execution**	

($\epsilon_{tol} \leq 3\,mm$) experience collapse at spatial displacements an order of magnitude smaller than domestic tasks in D3.

- **Hypothesis 3 (H_3: Error Compounding in Multi-Subtask Chains)**: *Chained compound prompt domains (D6 and D7) exhibit compounded failure cascades where small contact deviations in step 1 propagate exponentially into terminal failures in subsequent steps, dropping PRI below 15%.*

B. Preliminary Empirical Calibration

Table II reports our Stage 1 verification baselines. Evaluating the deterministic algorithmic oracles across disjoint test seeds confirms 100% nominal solvability across all 210 tasks, validating that the physical kinematic structures, tolerances, and contact predicates are fully feasible. Conversely, evaluating uniform random policies yields 0.0% success across all domains, verifying that tasks cannot be solved via accidental contact or trivial drift.

VIII. DISCUSSION AND STAGE 2 EXTENSION CHARTER ROADMAP

In accordance with the RSS 2027 Shepherding Process, this Stage 1 Extended Abstract establishes the foundational methodology, task taxonomy, qualitative rendering suite, training/evaluation protocols, and affordance perturbation manifold of AFFORD-Bench. For the Stage 2 final submission, we outline our planned Extension Charter: (1) *Full Empirical Scaling Across All 210 Tasks*: Executing complete evaluation matrices (50 rollouts per condition $\times$ 210 tasks across 35 benchmark suites) across Diffusion Policy, ACT, OpenVLA, SmolVLA, AtomVLA, and DP3; (2) *Disentangled Perturbation Ablations*: Isolating the independent effects of $\mathcal{P}_{\text{spatial}}$, $\mathcal{P}_{\text{physics}}$, and $\mathcal{P}_{\text{visual}}$ to map individual sensitivity frontiers; and (3) *Physical-to-Real Sim-to-Real Pilot*: Evaluating policy transfer onto a physical Franka Emika Panda workcell for representative D1 (gear meshing) and D4 (pipette docking) tasks. By uniting high physical precision, diverse real-world domains, rigorous protocols, and affordance-grounded perturbations, AFFORD-Bench provides the robotics community with an uncompromising benchmark to steer policy learning toward genuine physical generalizability.

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